Zainab Fayyaz

Junior Data Engineer | Python, SQL, Spark, Kafka, dbt, Airflow

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PROFESSIONAL SUMMARY

Computer Science graduate (NUML Lahore, 2026) transitioning from full-stack .NET development into data engineering, with four completed pipeline projects on real public data: a dbt star-schema warehouse, an AWS and Spark data lake, a Kafka and Spark streaming pipeline, and an orchestrated platform with tests and monitoring. Each project documents its design decisions, validation checks and known limitations. Previously built C# and ASP.NET Core applications and a Python machine-learning analytics project, with experience on both the application and data sides.

EDUCATION

BS Computer Science, National University of Modern Languages (NUML), Lahore 2022 – 2026
Coursework complete; final transcript pending

Intermediate in Computer Science, Punjab Group of Colleges 2020 – 2022 | 89%

TECHNICAL SKILLS

Data Engineering: Apache Kafka, Apache Spark (PySpark, Structured Streaming), dbt Core, Apache Airflow, Parquet, star-schema modelling, data quality testing, pipeline monitoring

Cloud and Databases: AWS (S3, IAM, Athena), Azure Blob Storage, PostgreSQL, SQL Server, MySQL

Languages and Tools: Python, SQL, C#, JavaScript, Docker, Docker Compose, Git, GitHub

Application Development: ASP.NET Core, .NET 8, Entity Framework Core, REST APIs, React, FastAPI, scikit-learn

KEY PROJECTS

Wikipedia Attention Platform | Kafka, Spark, PostgreSQL, dbt, Airflow, Docker GitHub

- Built a platform joining a Wikipedia edit stream (501,604 events through Kafka) with 24 hourly pageview files (49.5M English rows, Spark to Parquet) in PostgreSQL; dbt models with 18 passing tests show edited pages receive about 10 times the views of the average page.
- Orchestrated the platform with Airflow: a health-report task records row counts and match rate per run and fails the run on any problem, verified with a deliberate failure drill.
- Found and fixed a silent data-loss bug (time-based resume produced 174k instead of 501k events) by resuming from the stream's own position; documented limits in a decision log and runbook.

Wikipedia Real-Time Streaming Pipeline | Kafka, Spark Structured Streaming, PostgreSQL, Docker GitHub

- Streamed Wikipedia's live change feed (30 to 40 events per second) through Kafka into one-minute event-time windows with a 2-minute watermark in Spark, saved to PostgreSQL with checkpointing; a restart test resumed at the next batch with no duplicates.
- Validated end to end: 0 duplicates in 16,634 events checked, and Kafka and PostgreSQL counts reconciled, with the 4,915-event gap fully explained by three still-open windows.

NYC Taxi Cloud Data Lake | AWS S3, Athena, Spark, Azure Blob, Docker GitHub

- Built a bronze, silver and gold data lake on 48.7M NYC taxi trips: Spark flags 7 quality problems instead of deleting rows, and the gold table reconciles exactly (48,722,602 minus 2,264 excluded trips equals 48,720,338).
- Measured Athena scan cost: a partition filter scanned 6.14 MB against 86.68 MB for the same column unfiltered, about 14 times less.

Olist E-commerce Data Warehouse | dbt, PostgreSQL, Airflow, Docker GitHub

- Modelled about 100,000 orders from 8 CSV files into a star schema with 4 fact tables using dbt, with 32 passing tests and row counts reconciled to staging (99,441 orders, 112,650 items).
- Orchestrated load, dbt run and dbt test in Airflow, started with a single `docker compose up`.

SafeVision: Spatial Visualytics of Reported Incidents in Lahore | Python, FastAPI, React, MySQL

- Trained a Random Forest classifier on 25,440 FIR records across 31 areas and 18 crime categories (5-fold cross-validation) and served predictions through FastAPI, React and Leaflet dashboards.
- Built an Urdu FIR processing pipeline (Gemini 2.5 Flash, EasyOCR, Tesseract, OpenCV) with fuzzy matching against 265 Lahore locations.

PROFESSIONAL EXPERIENCE

Full Stack Web Developer Intern | SimpleLogix, Lahore Jul 2025 – Oct 2025

- Built and maintained full-stack web applications with .NET Framework, C#, Entity Framework, JavaScript and SQL Server, and developed REST APIs connecting the frontend and backend.
- Implemented database integration and CRUD operations with Entity Framework and SQL Server; participated in code reviews, debugging, performance improvements and Scrum.

CERTIFICATIONS

HEC National Skills Competency Test (NSCT), 88.1 percentile nationwide (Higher Education Commission, with MoITT, PSEB and P@SHA) 2026